

Date	3 JULY 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Demonstration of Proposed Features

Introduction

The **Credit Card Approval Prediction using Machine Learning** project was developed to demonstrate how machine learning techniques can assist financial institutions in identifying fraudulent credit card transactions efficiently. The application integrates a trained machine learning model with a Flask-based web interface, allowing users to enter transaction details and receive prediction results instantly. During the project demonstration, all major features of the application were presented to showcase the complete workflow from user input to prediction output.

User-Friendly Web Interface

The application provides a simple, responsive, and easy-to-use web interface developed using HTML and CSS. Users can access the application through a web browser and enter the required transaction details, including the **Time**, **Amount**, and anonymized transaction features **V1 to V28**. The interface is designed to be intuitive, ensuring that users can interact with the system without requiring technical knowledge.

Input Validation and Data Preprocessing

After the transaction details are entered, the application performs input validation to ensure that all required fields contain valid numerical values. The **Time** and **Amount** attributes are then processed using the trained feature scaling models to match the format used during machine learning model training. This preprocessing step improves prediction consistency and ensures that the input data is suitable for the Random Forest classifier.

Machine Learning-Based Prediction

The core functionality of the application is powered by a trained **Random Forest Classifier**, which was selected after comparing multiple machine learning algorithms, including Logistic Regression and Decision Tree. Once preprocessing is completed, the application loads the trained model and predicts whether the transaction is **Genuine** or **Fraudulent**. The prediction process is completed within a few seconds, providing users with fast and reliable results.

Result Display

The prediction generated by the machine learning model is displayed on a dedicated result page. The application clearly informs the user whether the entered transaction is classified as **Genuine** or **Fraudulent**. The clean presentation of the result improves readability and allows users to quickly understand the prediction without additional interpretation.

Performance and Reliability

The application was tested using multiple sample transactions to verify its functionality and prediction capability. Model performance was evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. Among the evaluated models, the Random Forest classifier demonstrated the

best overall performance and was therefore integrated into the final application. The modular architecture also ensures reliable execution and makes future enhancements easier to implement.

Future Enhancement Demonstration

The demonstration also discussed several possible improvements that can further enhance the application. These include cloud deployment for online accessibility, database integration for storing prediction history, user authentication for secure access, REST API development for system integration, mobile application support, and the incorporation of Explainable AI techniques to provide users with detailed explanations of prediction results.

Conclusion

The demonstration successfully showcased the complete functionality of the Credit Card Approval Prediction System, including user interaction, data preprocessing, machine learning prediction, and result generation. The application effectively illustrates how machine learning can be integrated into a web-based solution to automate fraud detection, improve prediction accuracy, and support faster decision-making in financial systems. The developed system is scalable, user-friendly, and serves as a strong foundation for future enhancements and real-world deployment.